



High-Resolution Imagery for Disaster Mapping

Drone Imagery Processing and Photogrammetric Workflows

Module 04 - Topic #2

Instructor: Dr. Giorgos SFIKAS (Assistant Professor, University
of West Attica)

DIGIHAZ is implemented by:



University of Alicante
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Learning Objectives

By the end of this lecture you will be able to:

- 01 Understand the basics of photogrammetric pipelines
- 02 Learn how to generate 2D Geospatial Products
- 03 Understand the basics of DSM and DEMs.

Topics Covered

- OpenDroneMap Ecosystem and Infrastructure
- Photogrammetric basics
- Digital Surface Models and Digital Terrain Models
- Calculating Building Heights and Structural Metrics

Estimated Duration
90 minutes lecture + 45 min exercise

Drone Imagery Processing and Photogrammetric Workflows

OpenDroneMap Ecosystem and Infrastructure



Drone Imagery Processing and Photogrammetric Workflows

OpenDroneMap Ecosystem and Infrastructure

Photogrammetric software

-  [Agisoft Metashape](#) - Process digital images and generate 3D spatial data
-  [Beholder Desktop](#) - Easy to use photogrammetry software, free for personal, non-commercial use. Also available as a SaaS.
-  [Meshroom](#)  Star  13k - A user-friendly 3D Computer Vision software
-  [MicMac](#)  Star  869 - A free open-source photogrammetric suite that can be used in a variety of 3D reconstruction scenarios
-  [MVE](#)  Star  1.1k - Includes Structure from Motion, Multi-View Stereo and Surface Reconstruction
-  [OpenDroneMap](#)  Star  5.9k - An open source command line toolkit for processing aerial drone imagery
-  [Pix4D](#) - A unique photogrammetry software suite for mobile and drone mapping
-  [PixPro](#) - Professional photogrammetry for everyone
-  [RealityCapture](#) - The state-of-the-art photogrammetry software solution that is changing the industry
-  [SURE by nFrames](#) - Software for 3D reconstruction from images with unique performance and flexibility
-  [TeleSculptor](#)  Star  643 - A cross-platform desktop application for photogrammetry
-  [Trimble Inpho](#) - Industry Leading Software for Digital Photogrammetry

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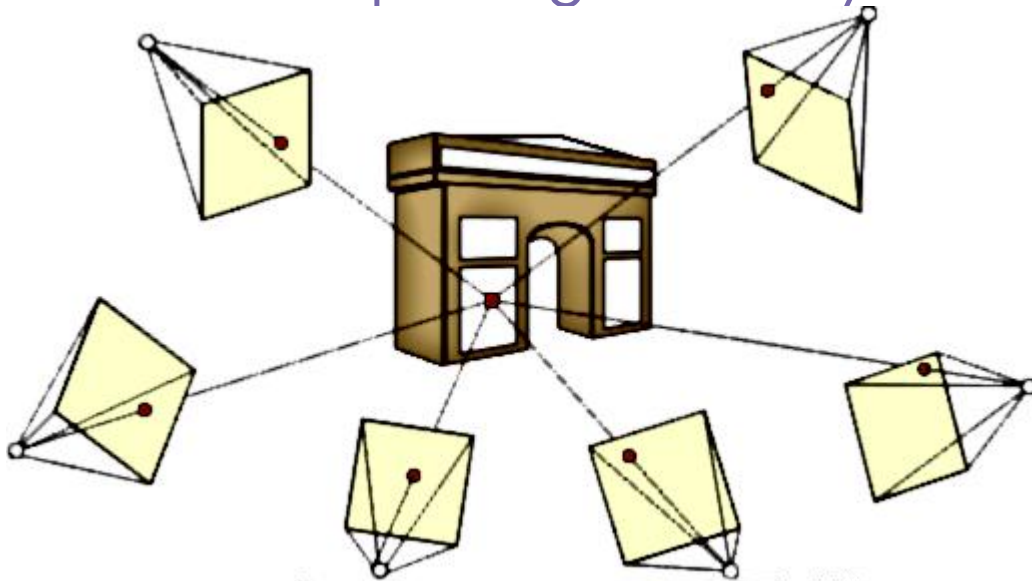
OpenDroneMap Ecosystem and Infrastructure

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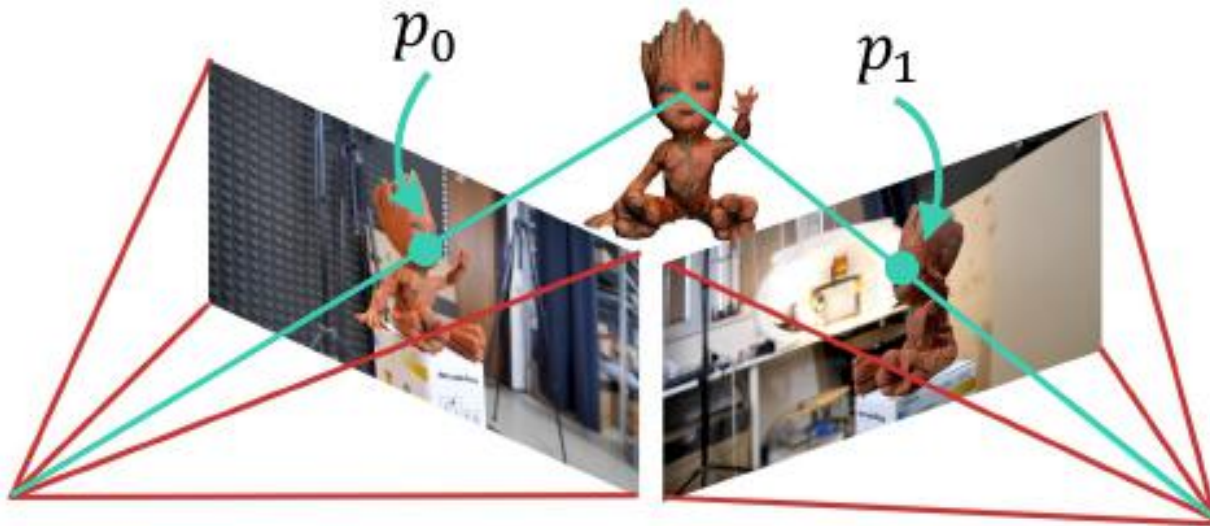
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A crash course on photogrammetry



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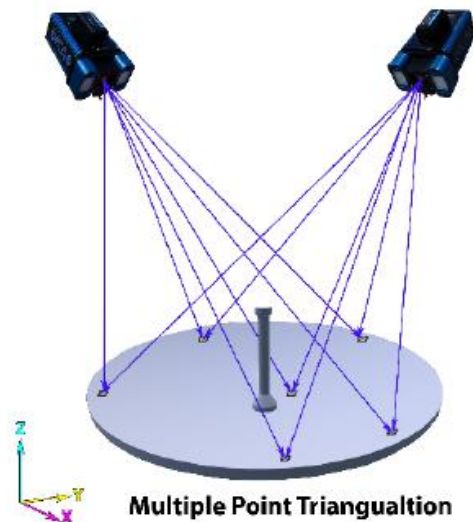
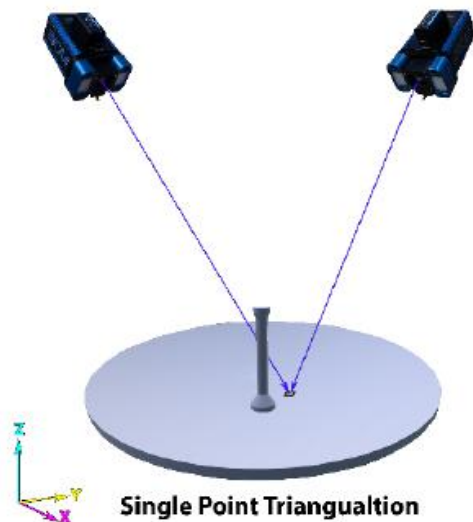
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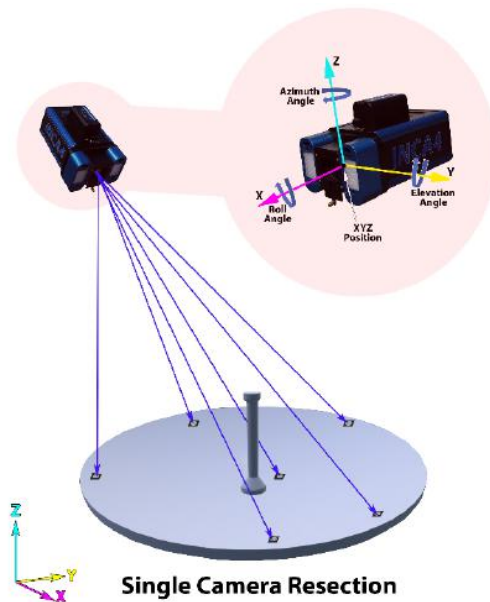
Triangulation



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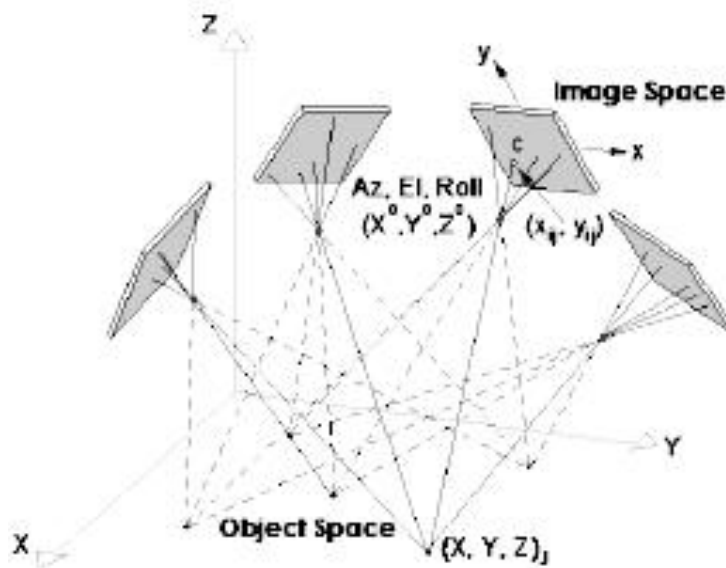
Camera resectioning



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Bundle adjustment



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The collinearity condition

- x, y : 2D coordinates
- X, Y, Z : 3D coordinates

$$x = x_o - c \frac{r_{11}(X - X_o) + r_{12}(Y - Y_o) + r_{13}(Z - Z_o)}{r_{31}(X - X_o) + r_{32}(Y - Y_o) + r_{33}(Z - Z_o)}$$

- Interior orientation: c, x_o, y_o

$$y = y_o - c \frac{r_{21}(X - X_o) + r_{22}(Y - Y_o) + r_{23}(Z - Z_o)}{r_{31}(X - X_o) + r_{32}(Y - Y_o) + r_{33}(Z - Z_o)}$$

- Exterior orientation: r, X_o, Y_o, Z_o

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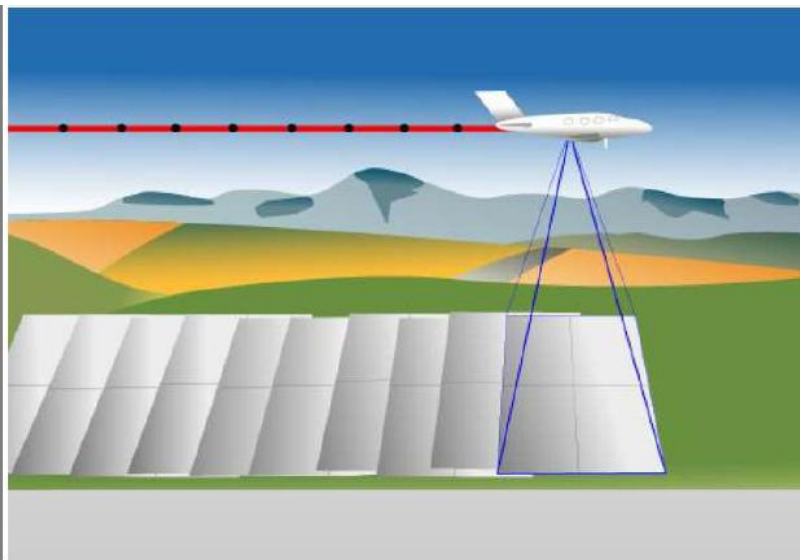
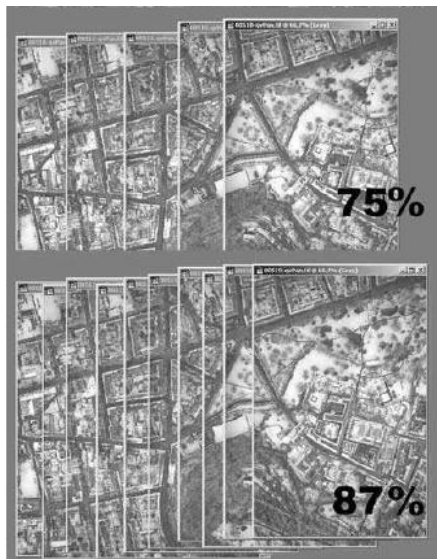
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dense depth map

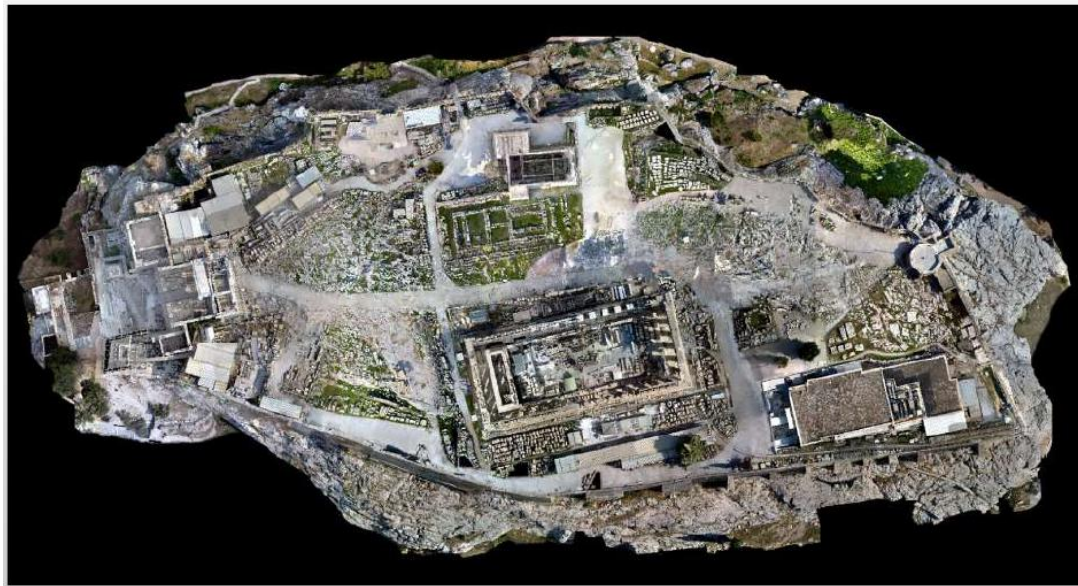
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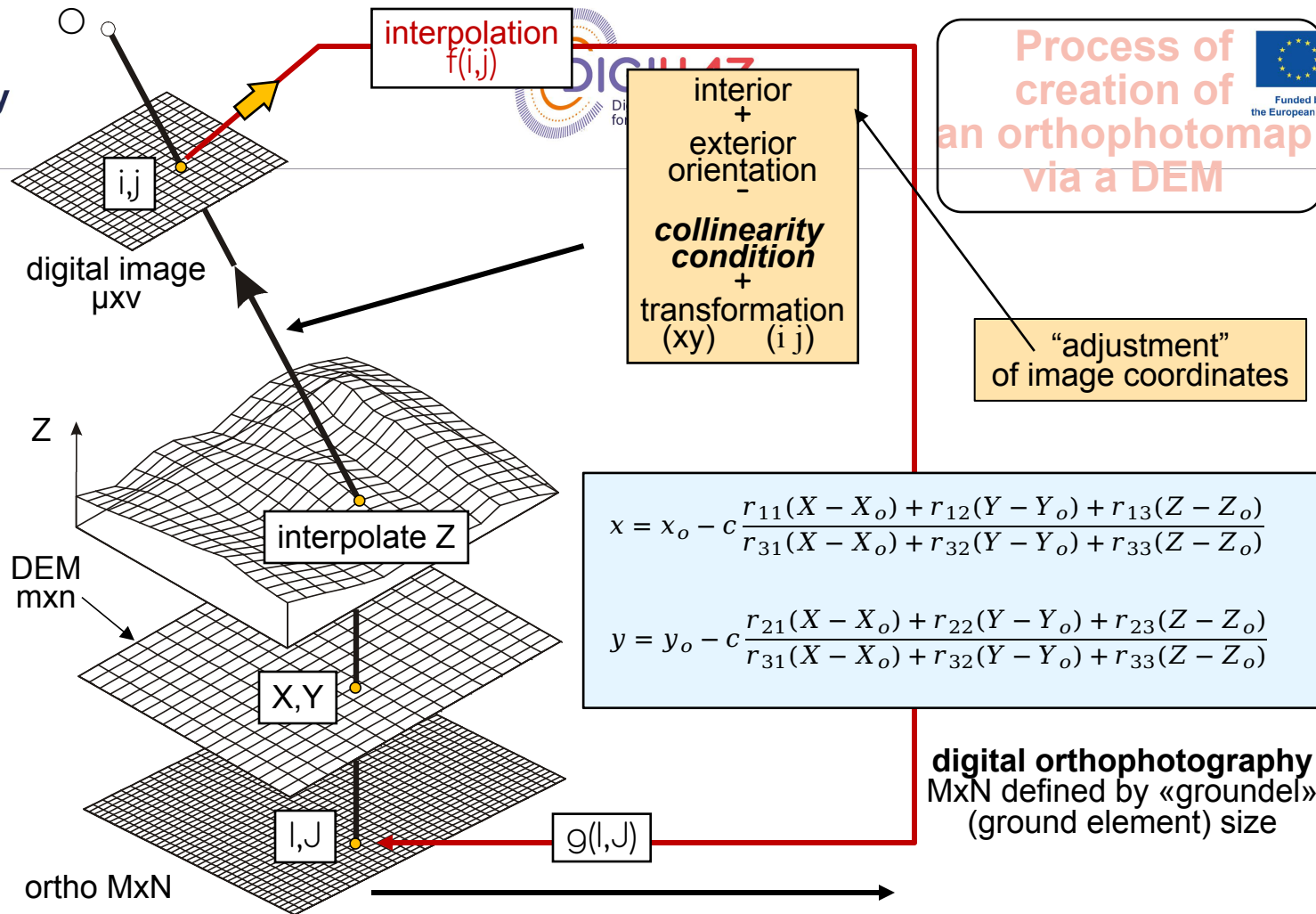


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Process of creation of an orthophotomap via a DEM



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Digital Surface Models and Digital Terrain Models

- Digital Elevation Model: 3D elevation data (Z) over a canonical grid (XY).

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Digital Surface Models and Digital Terrain Models

- Digital Elevation Model: 3D elevation data (Z) over a canonical grid (XY).
- In ODM: Run it with the «--dsm» flag: This creates a DSM
 - DSM -- the absolute elevation of the highest surfaces in the scanned area

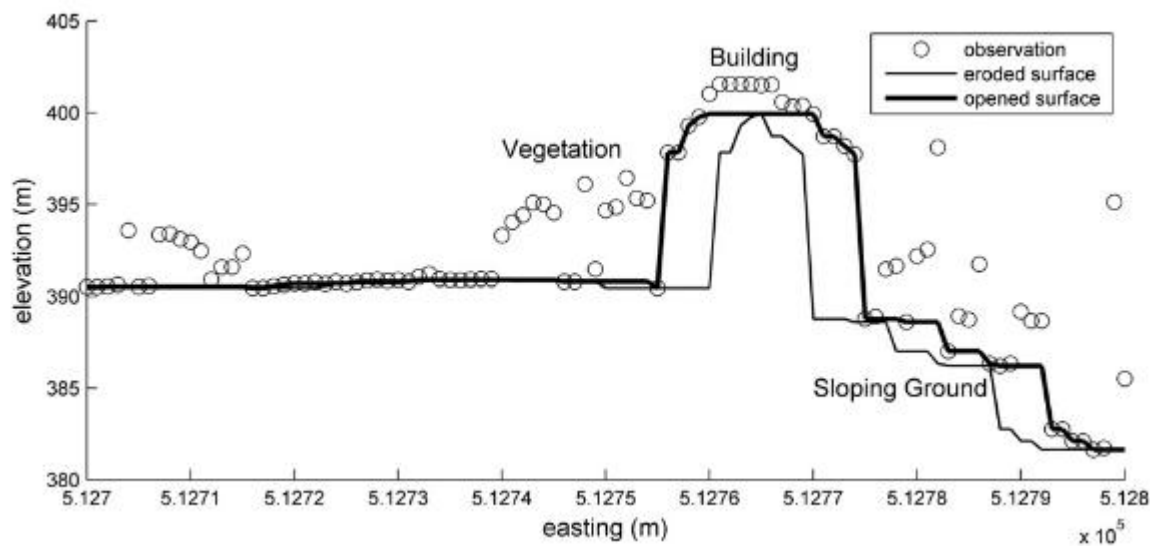
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Digital Surface Models and Digital Terrain Models

- Digital Elevation Model: 3D elevation data (Z) over a canonical grid (XY).
- In ODM: Run it with the «--dsm» flag: This creates a DSM
 - DSM -- the absolute elevation of the highest surfaces in the scanned area
 - DTM -- the «bare earth» elevation

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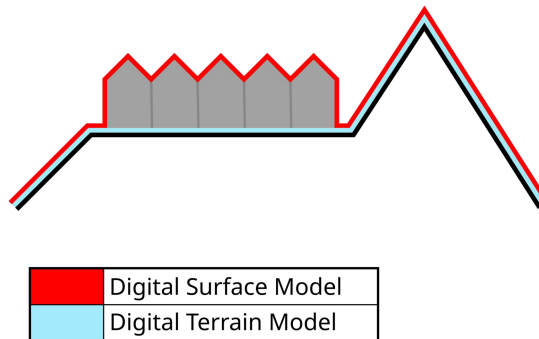
Digital Surface Models and Digital Terrain Models



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Calculating Building Heights and Structural Metrics

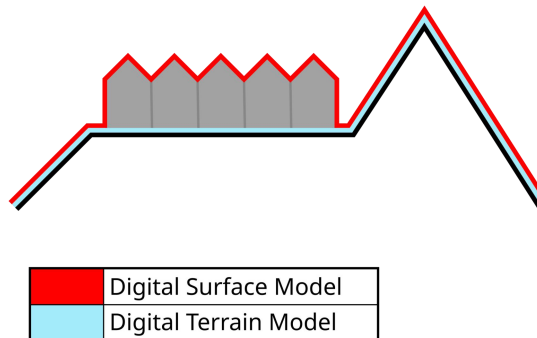
- The relation of DSM and DTM forms the basis of estimating structure heights.



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Calculating Building Heights and Structural Metrics

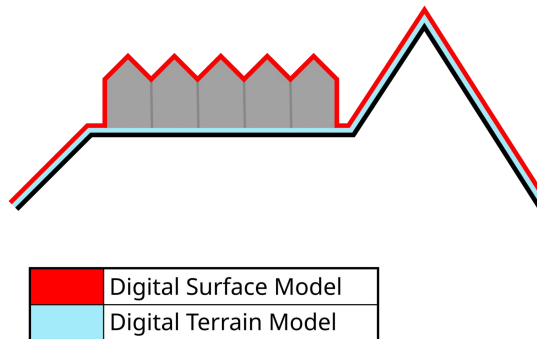
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- $nDSM = DSM - DTM$



Drone Imagery Processing and Photogrammetric Workflows

Calculating Building Heights and Structural Metrics

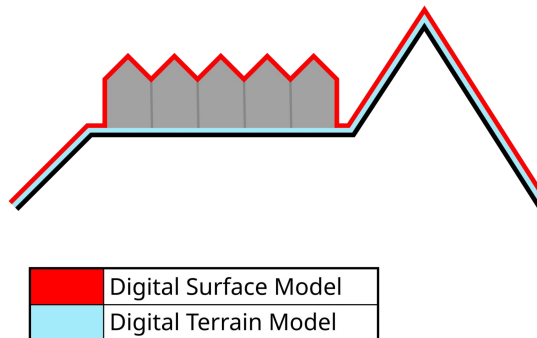
- The relation of DSM and DTM forms the basis of estimating structure heights.
- $nDSM = DSM - DTM$
- Easily visualize in GIS



Drone Imagery Processing and Photogrammetric Workflows

Calculating Building Heights and Structural Metrics

- The relation of DSM and DTM forms the basis of estimating structure heights.
- $nDSM = DSM - DTM$
- Easily visualize in GIS
- In a rapid disaster response context: Used to flag areas that suffered «pancaking», roof collapse, etc.



Key terms & definitions

Key Terms & Definitions	
Photogrammetry	The art, science, and technology of obtaining reliable information about physical objects and the environment, through processes of recording, measuring, and interpreting images and patterns of electromagnetic radiant energy and other phenomena[1]. These processes generate estimates about the spatial environment, such as 3D models and orthophotomaps, from a series of 2D photographs.
Aerophotogrammetry	Photogrammetry with an input of aerial imagery.
Structure from Motion (SfM)	The photogrammetric process, in the sense of using 2D photographs as inputs and setting as a goal to perform estimates about the 3D environment. In SfM, these are in particular the exact camera positions corresponding to the input images, and the 3D geometry of the photographed scene.

Key terms & definitions

Key Terms & Definitions	
Orthomosaic	A highly detailed, geometrically corrected aerial photograph (or a stitched composite of many photographs) where perspective distortion has been removed so that the scale is uniform across the entire image, allowing it to be used exactly like a map.
Digital Elevation Model (DEM)	An umbrella term for a three-dimensional representation of terrain or surfaces, which broadly encompasses both Digital Surface Models (DSMs) and Digital Terrain Models (DTMs).
Digital Surface Model (DSM)	A 3D elevation model that captures the absolute height of everything sitting on top of the earth's surface, encompassing buildings, tree canopies, infrastructure, and the terrain itself.
Digital Terrain Model (DTM)	A 3D elevation model representing the bare-earth surface, generated by algorithmically stripping away all above-ground features such as vegetation and human-made structures from the dataset.

Unmanned Systems & Image Processing Basics

Discussion questions

1. In your own words, how would you define the photogrammetric process, and how do software like ODM address the end-goals of photogrammetry?
2. Define & explain similarities and differences of the following notions: Digital Elevation Model, Digital Terrain Model, Digital Surface Model.
3. Explain the steps you would take in a real-world scenario where building height estimation is required.

Sneak peak at your exercise

Example

Running OpenDroneMap on a
collection of images

See also:

GitHub repo / Notebook link

```
// Place your images (e.g. jpegs) in a folder named "images" (for  
example /my/project/images).  
// A recommended set is material from the "odm_mygla_dataset".  
You can download it using:  
git clone git@github.com:merkato/odm_mygla_dataset.git  
  
// After installing Docker, simply run:  
docker run -ti --rm -v /my/project:/datasets/code  
opendronemap/odm --project-path /datasets
```

Thank you!

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